

Contrastive Learning for Representation

Abstract

This project investigates how training and model-design choices affect the quality and structure of learned representations using contrastive learning on CIFAR-10. A convolutional encoder was trained using two augmented views of each image and an InfoNCE contrastive objective. Representation quality was evaluated using k-nearest-neighbor (k-NN) classification, representation geometry, PCA visualization, and nearest-neighbor retrieval.

The trained encoder achieved **48.76% k-NN accuracy**, compared with **33.19% for a randomly initialized encoder**, an improvement of 15.57 percentage points. The learned representation also produced a substantially larger same-class versus different-class cosine-similarity gap (**0.1584**) compared with the random encoder (**0.0020**). Retrieval performance improved from 29.10% to 44.60% Precision@1.

Three controlled experiments were performed to study temperature, augmentation strength, and projection-head design. The experiments demonstrate that representation quality depends strongly on these choices and that contrastive loss alone is not sufficient to judge representation quality.

1. Introduction

Representation learning aims to learn useful feature vectors from raw data such that samples with meaningful relationships are represented appropriately in feature space.

Contrastive learning provides a way to learn such representations without requiring the model to directly predict class labels during training. Instead, two augmented views of the same image are treated as a positive pair, while views originating from different images form negative examples.

The central research question of this project is:

How do different training and model-design choices affect the quality and structure of learned representations?

To investigate this question, a contrastive learning pipeline was implemented on CIFAR-10 and evaluated using both quantitative and qualitative measures. In addition to a baseline trained model, controlled experiments were performed by changing temperature, augmentation strength, and projection-head usage.

2. Dataset and Preprocessing

The experiments use the CIFAR-10 dataset, containing 50,000 training images and 10,000 test images across 10 classes:

airplane, automobile, bird ,cat,deer ,dog, frog, horse ,ship, truck

Images are RGB and have spatial dimensions of 32×32 pixels.

Images were normalized using the standard CIFAR-10 channel statistics. During contrastive training, each image was independently transformed twice to produce two different views of the same underlying image.

Two augmentation settings were investigated:

Weak augmentation

- Random horizontal flip
- Normalization

Strong augmentation

- Random resized crop
- Random horizontal flip
- Color jitter
- Random grayscale
- Normalization

The two resulting views form a positive pair.

3. Contrastive Learning Method

3.1 Positive and Negative Pairs

For an image x_i , two independently augmented views are generated:

$$x_i(1) = t_1(x_i), x_i(2) = t_2(x_i)$$

These form a positive pair.

Other images in the minibatch provide negative examples. Thus, the model learns to produce similar representations for different views of the same image while distinguishing representations originating from different images.

3.2 Encoder

The encoder is a convolutional neural network consisting of four convolutional blocks with batch normalization and ReLU activations.

The spatial representation is reduced using strided convolutions and adaptive average pooling. A fully connected layer produces a **128-dimensional representation**:

$$h \in R^{128}$$

This encoder representation is used for downstream evaluation.

3.3 Projection Head

The main model uses a two-layer projection head:

$$h \in R^{128} \rightarrow R^{128} \rightarrow z \in R^{64}$$

The projection output is L2-normalized before calculating contrastive similarity.

The projection head separates the representation used for the contrastive objective from the representation used for downstream evaluation.

A controlled experiment also compares the model with and without this projection head.

3.4 InfoNCE Loss

The contrastive objective uses the normalized embeddings and cosine similarity. For each positive pair, the model increases the similarity between the positive examples while decreasing similarity to negative examples.

A temperature parameter controls the sharpness of the similarity distribution. The main model uses:

$$T=0.5, T = 0.5$$

The temperature was also varied experimentally using $T = 0.1$, $T=0.1$, $0.50.5$, and $1.01.0$.

4. Training Setup

The main model was trained for three epochs using:

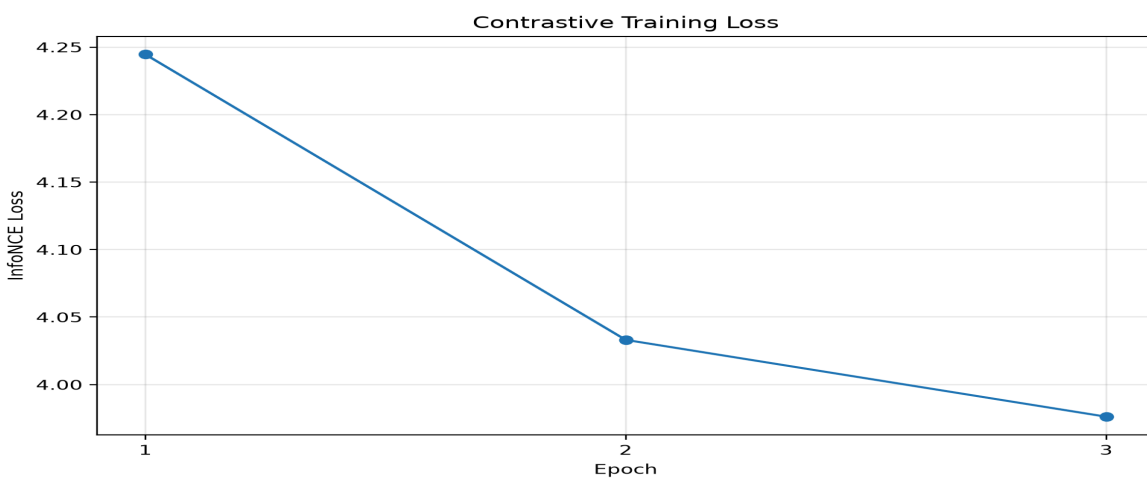
Parameter	Value
Dataset	CIFAR-10

Batch size	128
Learning rate	0.0003
Weight decay	0.0001
Temperature	0.5
Encoder dimension	128
Projection dimension	64
Augmentation	Strong
Random seed	42
Device	CPU

The contrastive loss decreased throughout training:

Epoch	InfoNCE Loss
1	4.2446
2	4.0331
3	3.9762

Figure1: Training loss



The decreasing loss indicates that the model successfully optimized the contrastive objective during training.

5. Representation Evaluation

Representation quality was evaluated using four complementary approaches:

1. k-NN classification
2. Representation geometry
3. PCA visualization
4. Nearest-neighbor retrieval

A randomly initialized encoder was used as the baseline.

5.1 k-NN Classification

The encoder representation h_h was extracted from the training and test images. A 5-nearest-neighbor classifier was then used to evaluate whether the learned representations contain useful class information.

Encoder	k-NN Accuracy
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Random	33.19%
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Trained	48.76%
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The trained representation improved k-NN accuracy by **15.57 percentage points**, demonstrating that contrastive training produced representations containing more useful class-related information than the random baseline.

Class-wise accuracy varied across CIFAR-10 categories. For example, frog achieved 60.00%, while cat achieved 29.10%, indicating that some classes remained substantially more difficult to separate.

5.2 Representation Geometry

Cosine similarity was used to compare samples from the same class and different classes.

Metric	Random	Trained
Same-class similarity	0.9781	0.5217
Different-class similarity	0.9761	0.3632
Similarity gap	0.0020	0.1584
Average embedding variance	0.00001767	0.09100722

The random encoder produced extremely high similarity for both same-class and different-class samples. Its similarity gap was only 0.001969.

After contrastive training, the same-class/different-class gap increased to 0.158434.

This indicates that training changed the geometry of the representation space so that similarity became more informative about class relationships.

The very low variance of the random embeddings also indicates that the random representation was highly concentrated.

5.3 PCA Visualization

PCA was applied to 2,000 embeddings for visualization.

The first two principal components explained:

Encoder	PC1	PC2	PC1 + PC2
Random	54.42%	24.69%	79.11%
Trained	38.65%	15.95%	54.61%

The random representation has higher explained variance in its first two principal components. However, explained variance alone is not a measure of representation quality.

The geometry and k-NN results provide stronger evidence that the trained representation contains more useful class-sensitive structure.

Figure 2. PCA visualization of random encoder representations:

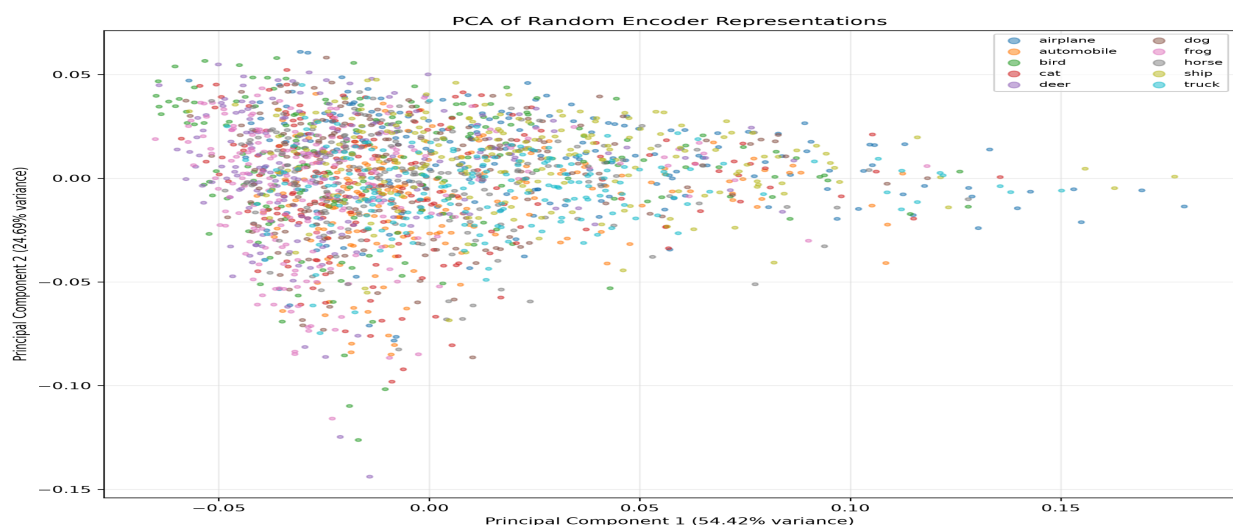
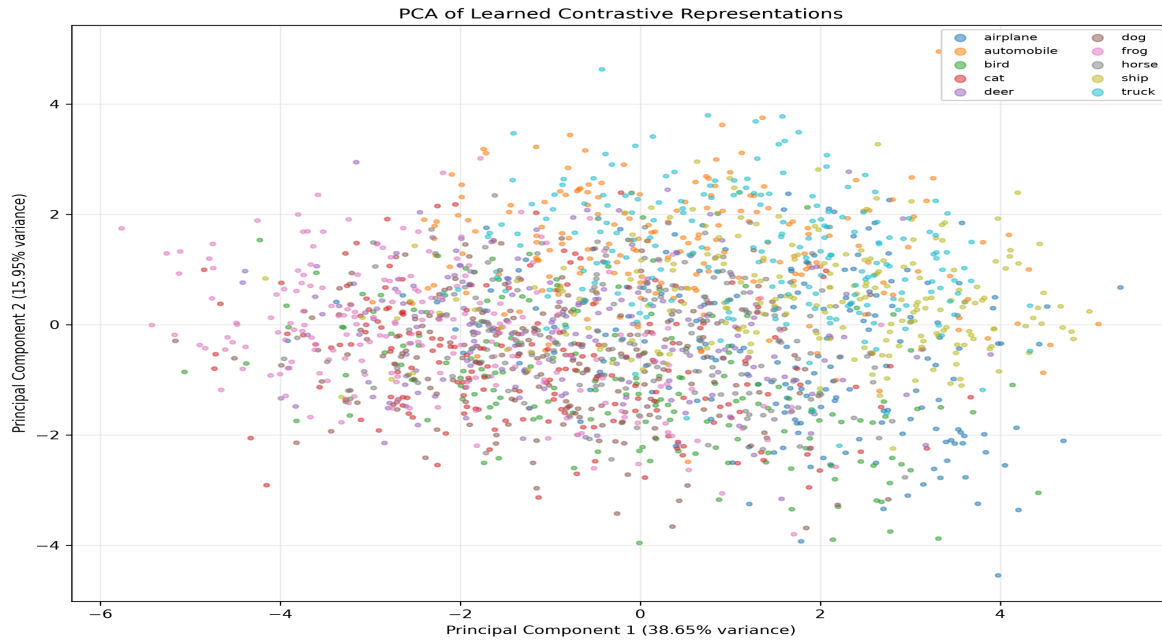


Figure 3. PCA visualization of learned encoder representations:



6. Retrieval Evaluation

The learned embeddings were also evaluated using nearest-neighbor image retrieval.

The test set was used as the retrieval database, with 1,000 queries evaluated using cosine similarity.

Metric	Random	Trained
Precision@1	29.10%	44.60%
Precision@5	25.22%	38.46%
Accuracy@1	29.10%	44.60%
Accuracy@5	68.60%	78.30%

The trained representation consistently outperformed the random baseline.

This demonstrates that the learned representation is useful not only for k-NN classification but also for similarity-based retrieval.

Figure 4. Example nearest-neighbor retrieval using the random encoder:

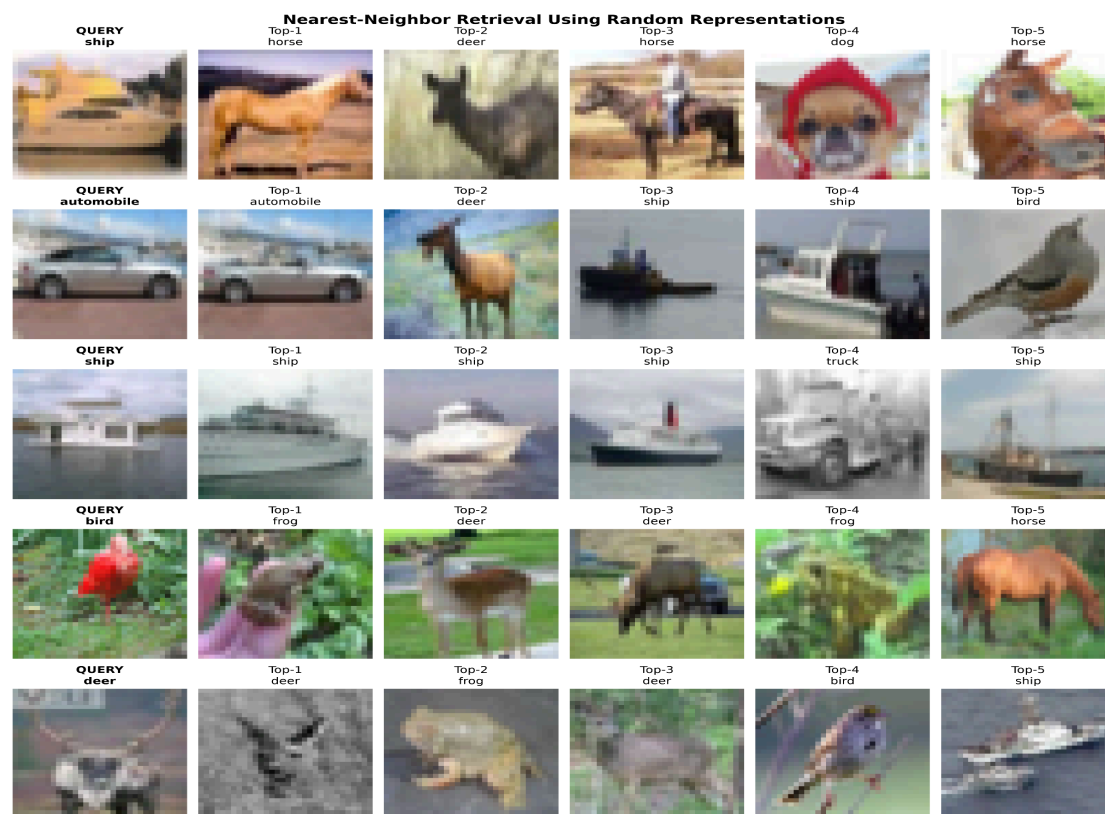
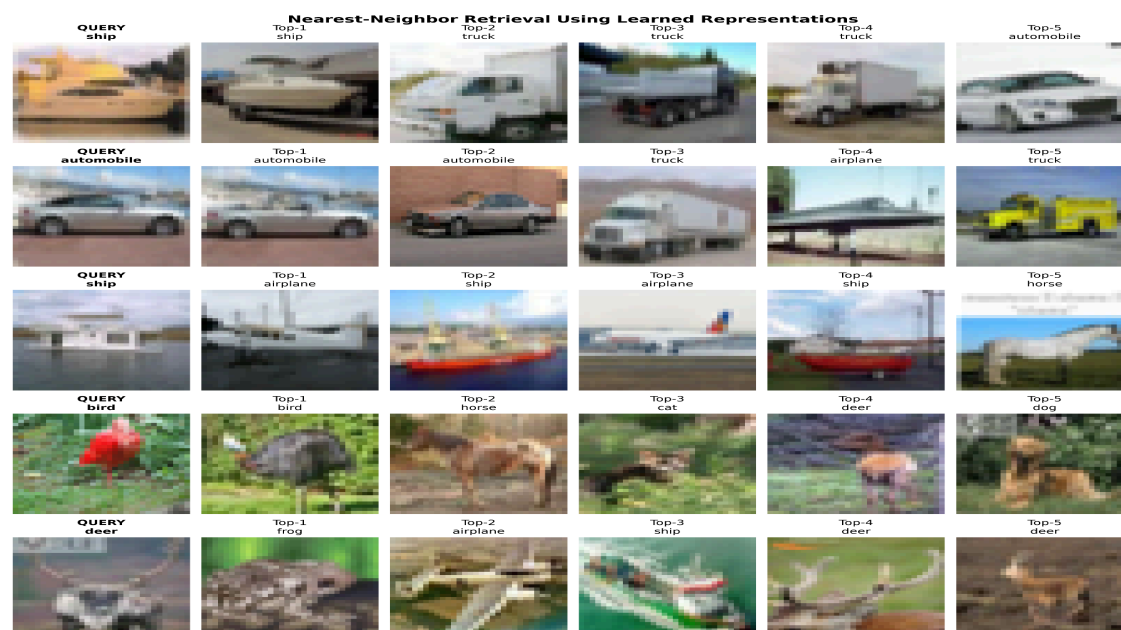


Figure 5. Example nearest-neighbor retrieval using the trained encoder:



7. Controlled Experiments

To determine how training and architecture choices affect the representation, three controlled experiments were conducted. In each experiment, one factor was varied while the other main training settings were kept fixed.

7.1 Temperature

Three temperature values were tested.

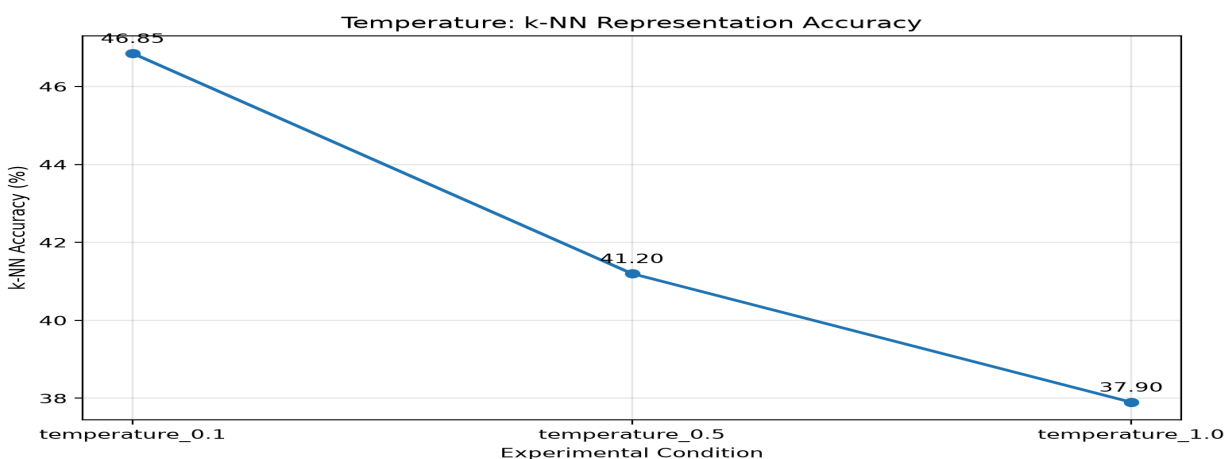
Temperature	Final Loss	k-NN	Similarity Gap	P@1	P@5
0.1	0.8833	46.85%	0.0965	43.50%	39.06%
0.5	4.0331	41.20%	0.1244	39.80%	36.14%
1.0	4.7397	37.90%	0.1113	37.30%	33.98%

Temperature clearly affected both optimization behavior and representation quality.

The lowest temperature tested produced the strongest k-NN and retrieval results in this experiment, while $T=0.5$ produced the largest similarity gap.

Importantly, loss values at different temperatures should not be directly interpreted as representation-quality rankings because temperature changes the scale and sharpness of the similarity logits.

Figure 6. k-NN accuracy across temperature settings:



7.2 Augmentation Strength

Weak and strong augmentation were compared.

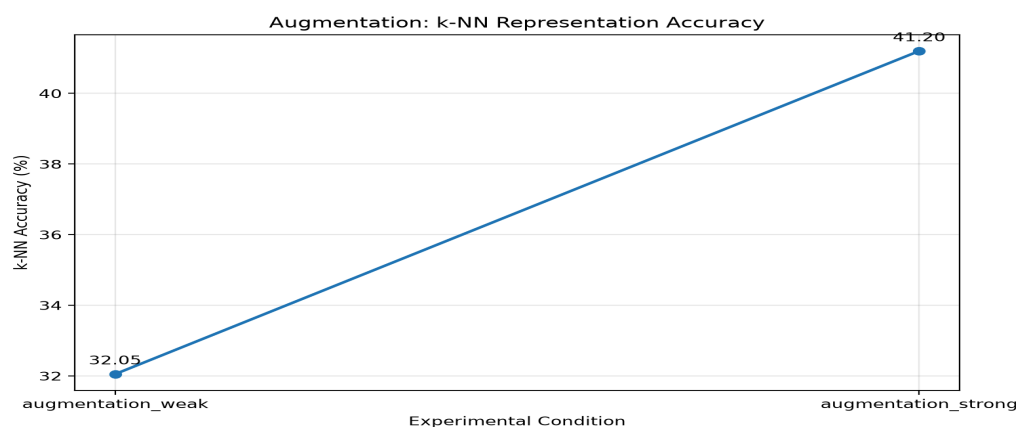
Augmentation	Final Loss	k-NN	Similarity Gap	P@1	P@5
Weak	3.6245	32.05%	0.0584	27.10%	25.48%
Strong	4.0331	41.20%	0.1244	39.80%	36.14%

Strong augmentation produced a higher contrastive loss but substantially better downstream representation quality.

The stronger transformations make the positive-pair matching problem more difficult. Solving this harder task encourages the encoder to learn representations that remain useful despite larger changes between views.

This demonstrates why contrastive loss should not be considered in isolation when evaluating representation quality.

Figure .7 k-NN accuracy for weak and strong augmentation:



7.3 Projection Head

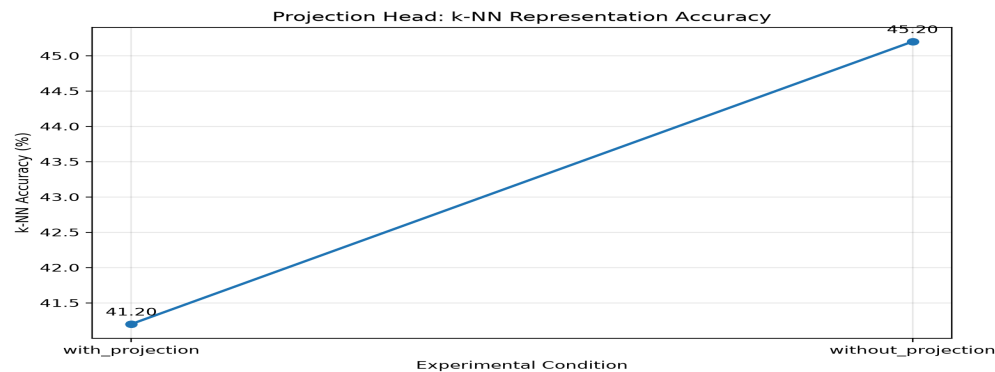
The effect of the projection head was investigated by comparing models with and without the projection head.

Configuration	Final Loss	k-NN	Similarity Gap	P@1	P@5
With projection	4.0331	41.20%	0.1244	39.80%	36.14%
Without projection	4.0233	45.20%	0.1583	41.90%	37.86%

Under the two-epoch experimental budget, the model without the projection head produced stronger downstream encoder metrics.

This should not be interpreted as evidence that projection heads are inherently harmful. The result may depend on the architecture, optimization settings, and training duration. Longer training and systematic variation of projection dimensions would provide a stronger test.

Figure .8 k-NN accuracy with and without the projection head:



8. Failure and Unexpected Results

8.1 Strong Augmentation Produced Higher Loss

What happened: Strong augmentation produced higher contrastive loss than weak augmentation but substantially better k-NN and retrieval performance.

Why: Stronger transformations create a more difficult positive-pair matching problem while encouraging greater invariance.

Possible improvement: Test several intermediate augmentation strengths and train for more epochs.

8.2 Removing the Projection Head Performed Better

What happened: The no-projection model achieved better downstream metrics under the controlled experimental budget.

Why: The projection head adds an additional transformation and parameters to the optimization process. The short training schedule may not have been sufficient for its benefits to emerge.

Possible improvement: Increase training duration and evaluate multiple projection-head dimensions and architectures.

8.3 Random Encoder Produced Very High Similarity

What happened: The random encoder had an average pairwise cosine similarity of approximately 0.976.

Why: The embeddings were highly concentrated. Consequently, both same-class and different-class samples appeared similar.

Possible improvement: Train the encoder with the contrastive objective so that the representation space develops meaningful differences between samples.

This failure case demonstrates why high similarity alone is not evidence of a good representation. The **difference between relevant and irrelevant similarities** is more informative.

9. Discussion

The experiments demonstrate that contrastive learning substantially changes the structure and usefulness of learned representations.

The trained encoder improved k-NN accuracy from 33.19% to 48.76% and increased the same-class versus different-class similarity gap from 0.001969 to 0.158434. Retrieval performance also improved substantially.

The controlled experiments demonstrate that representation quality is sensitive to training choices. Temperature affected both optimization and downstream performance. Strong augmentation improved representation quality despite increasing contrastive loss. The projection-head experiment produced an unexpected result in which removing the projection head performed better under the short experimental schedule.

These results also demonstrate that no single metric completely describes representation quality. Loss, similarity, embedding variance, classification accuracy, retrieval performance, and visualization provide complementary information.

10. Limitations

Several limitations should be considered.

First, the main model was trained for only three epochs, while the controlled experiments used two epochs. Longer training could change the relative performance of the tested configurations.

Second, the experiments were conducted on CIFAR-10 only, so the conclusions may not generalize directly to other datasets.

Third, the controlled experiments vary only a limited set of design choices. Additional experiments involving batch size, encoder capacity, learning rate, projection dimension, and augmentation parameters could provide a more comprehensive analysis.

Finally, PCA visualization is only a two-dimensional projection and should not be treated as a complete description of the 128-dimensional representation space.

11. Conclusion

This project investigated how training and model-design choices affect the quality and structure of contrastively learned representations on CIFAR-10.

The trained encoder produced substantially more useful representations than a random encoder. k-NN accuracy increased from **33.19% to 48.76%**, while the same-class versus different-class similarity gap increased from **0.001969 to 0.158434**. Retrieval Precision@1 increased from **29.10% to 44.60%**.

The controlled experiments showed that temperature, augmentation strength, and projection-head design all influence the resulting representation. Strong augmentation produced better downstream representations despite higher contrastive loss, while the no-projection configuration unexpectedly performed better under the limited experimental training budget.

Overall, the experiments demonstrate that evaluating contrastive learning requires examining both the optimization objective and the resulting representation space. Downstream performance, geometry, and retrieval provide complementary evidence of whether the learned representation is actually useful.

12. Reproducibility

The complete implementation is provided in the project repository.

The repository contains:

- Dataset and augmentation implementation
- Encoder and projection-head architecture
- InfoNCE loss
- Training pipeline
- Representation evaluation
- Retrieval evaluation
- Controlled experiments
- Experiment analysis
- Visualization scripts
- Generated experimental results

All main evaluation experiments use a fixed random seed of **42** to improve reproducibility.

Generated results and figures are stored under the [results/](#) directory, while experimental configurations and analysis are stored under [results/experiments/](#).